

Functional Requirements

1. UVSIM should be able to read BasicML programs.
2. UVSIM should be able to execute BasicML programs.
3. UVSIM should load programs from files into its memory.
4. UVSIM should have 100 words of memory.
5. UVSIM's CPU should have an accumulator.
6. UVSIM's CPU should have a control register.
7. UVSIM should store the address of the next instruction to be executed in the control register.
8. UVSIM should be able to take input from the keyboard.
9. UVSIM should be able to display output.
10. UVSIM should GUI.
11. UVSIM's GUI should have a file picker that allows users to choose the program they want to load.
12. UVSIM's GUI should have a button that opens the file picker.
13. UVSIM's GUI should have a button that begins the execution of the loaded program.
14. UVSIM's GUI should have a prompt window to take user input.
15. UVSIM's GUI should have a dialog window to display output.

Non-functional Requirements

1. Each component should be designed in a way that makes it easy to add new features without requiring modifications to other parts of the program.
2. The input of UVSIM's CPU should be decoupled from the CPU's implementation. (i.e., the CPU should be able to be used with any source of input.)
3. The output UVSIM's CPU should be decoupled from the CPU's implementation. (i.e., the CPU should be able to be used with any destination for output.)